THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET C

Chapters 3 & 4: **Metals and Non-metals, Carbon and its Compounds** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Bauxite is an ore of: 1

(a) zinc **(b)** iron **(c)** copper **(d)** aluminium

2. A student states: “Copper displaces silver from silver nitrate solution.” The correct response is: 1

(a) Correct **(b)** Incorrect
(c) Only on heating **(d)** Cannot be determined

3. Stainless steel is an alloy of iron with: 1

(a) zinc **(b)** tin
(c) lead **(d)** chromium and nickel

4. The number of valence electrons in a magnesium atom is: 1

(a) 8 **(b)** 3 **(c)** 2 **(d)** 1

5. For the anode in electrolytic refining of copper, Student A says impure copper; Student B says pure copper. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

6. Diamond is hard because: 1

(a) each C is bonded to four others in a 3-D network
(b) it has layers
(c) it is ionic
(d) it has free electrons

7. A student states: “Unsaturated fats are healthier than saturated fats.” The correct response is: 1

(a) Correct **(b)** Incorrect
(c) Both are equally harmful **(d)** Cannot be determined

8. Which compound has a triple bond? 1

(a) ethane **(b)** methane **(c)** ethyne **(d)** ethene

9. The product formed when ethanoic acid reacts with sodium carbonate includes: 1

(a) H_2 **(b)** CO_2 **(c)** CH_4 **(d)** O_2

10. For the soap molecule, Student A says its ionic end is hydrophilic; Student B says the hydrocarbon end is hydrophilic. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Write the equation for the reaction of aluminium oxide with sodium hydroxide. 2

12. Name two metals that do not react with water at all. 2

13. Write the electron-dot structure of cyclohexane in words (number of bonds). 2

14. Why is the conversion of ethanol to ethanoic acid an oxidation reaction? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. An element X (atomic number 12) reacts with element Y (atomic number 17). 3
(i) What type of bond is formed?
(ii) Write the formula.
(iii) Is the compound soluble in water?

16. Describe one test each for 3
(i) unsaturation
(ii) carboxylic acid
(iii) alcohol (with sodium).

17. Explain why copper vessels develop a green coating and silver articles turn black. How can such tarnish be removed from copper? 3

OR

What happens when a metal reacts with an acid? Write the equation of magnesium with dilute sulphuric acid. Name the gas and give its test. 3

18. What are the products of complete combustion of carbon compounds? Why are they used as fuels? Give one example. 3

OR

What is the role of a catalyst in hydrogenation of oils? Name the catalyst and the product. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** Old buildings have iron grills that rust, while temple bells made of bronze last centuries. 5
(i) What is rust?
(ii) Why do bronze bells not rust?
(iii) What is an alloy?
(iv) Name the constituents of bronze.
(v) State one advantage of alloying.

OR

(i) What are amphoteric oxides? 5
(ii) Write equations of ZnO with HCl and NaOH.
(iii) Name a neutral oxide of a non-metal.

20. **Case study:** Food is cooked in a gas stove. The bottom of a vessel sometimes becomes black. 5
(i) Why does this happen?
(ii) What does it indicate about the air holes?
(iii) What fuel is used in LPG?
(iv) Why is incomplete combustion dangerous?
(v) Write the equation of complete combustion of butane.

OR

(i) Differentiate between ethanol and ethanoic acid on the basis of 5
(a) litmus
(b) $NaHCO_3$
(c) smell.
(ii) Write the structures of both.

END OF QUESTION PAPER — **Total: 40 Marks**